about quantum mechanics. That's hardly unique, of course; scientists are always assuming that they can choose to measure whatever they want. But really we think that's just a convenient way of talking, and even those scientists are composed of particles and forces that themselves obey the laws of physics. So we can imagine invoking *superdeterminism*—the idea that the true laws of physics are utterly deterministic (no randomness anywhere), and furthermore that the initial conditions of the universe were laid down at the Big Bang in just precisely such a way that certain "choices" are never going to be made. It's conceivable that one could invent a perfectly local superdeterministic theory that would mimic the predictions of quantum entanglement, simply because the universe was prearranged to make it appear that way. This seems unpalatable to most physicists; if you can delicately arrange your theory to do that, it can basically be arranged to do anything you want, and at that point why are we even doing physics? But some smart people are pursuing the idea.

The other potentially doubtable assumption seems uncontroversial at first glance: that measurements have definite outcomes. When you observe the spin of a particle, you get an actual result, either spin-up or spin-down along whatever axis you are measuring it with respect to. Seems reasonable, doesn't it?

But wait. We actually know about a theory where measurements don't have definite outcomes—austere, Everettian quantum mechanics. There, it's simply not true that we get either up or down when we measure an electron's spin; in one branch of the wave function we get up, in the other we get down. The universe as a whole doesn't have any single outcome for that measurement; it has multiple ones. That doesn't mean that Bell's theorem is wrong in Many-Worlds; mathematical theorems are unambiguously right, given their assumptions. It just means that the theorem doesn't apply. Bell's result does not imply that we have to include spooky action at a distance in Everettian quantum mechanics, as it does for boring old single-world theories. The correlations don't come about because of any kind of influence being transmitted faster than light, but because of branching of the wave function into different worlds, in which correlated things happen.

For a researcher in the foundations of quantum mechanics, the relevance of Bell's theorem to your work depends on exactly what it is you're trying to do. If you have devoted yourself to the task of inventing a new version of